

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
(KUET)

Department of Computer Science and Engineering

3D Virtual Architectural Walkthrough: Khan Jahan Ali Hall

Course: CSE 4102

Computer Graphics and Image Processing Laboratory

SUBMITTED BY

Mehedi Hasan Rabby

Roll: 2107086

Year: 4th Term: I

Department of CSE

Khulna University of Engineering & Technology

SUPERVISED BY

Mr Md. Tajmilur Rahman

Lecturer, Dept of CSE, KUET

Mr Md. Mubtashim Abrar Nihal

Lecturer, Dept of CSE, KUET

Contents

1	Introduction and Project Scope	3
1.1	Project Motivation and Architectural Context	3
1.2	Proposed vs. Implemented Feature Matrix	3
2	Mathematical Computation of 3D Geometric Primitives	4
2.1	Primitive 1: The Unit Cube (Axis-Aligned Box)	4
2.2	Primitive 2: The UV Sphere (Latitude-Longitude Mesh)	5
2.3	Primitive 3: The Capped Cylinder	6
2.4	Primitive Geometry and Buffer Memory Footprint Analysis	7
2.5	Coordinate Transformation Pipeline	7
3	Hierarchical Modeling and Architectural Scene Construction	8
3.1	Interior Environment: Khan Jahan Ali Hall (Room 304)	8
3.2	Hallway Corridor Environment	8
3.3	Balcony Terrace and Quad Athletic Sports Complex	9
4	Illumination and Shading Models: Mathematical Formulation	9
4.1	Phong Reflection Model Components	9
4.2	Point Light Distance Attenuation	10
4.3	Multi-Source Illumination Infrastructure	10
4.4	Shading Model Comparison: Gouraud vs. Phong	10
4.5	Normal Matrix Transformation	10
5	Dynamic Kinematics, Animations, and Camera Navigation	11
5.1	Smooth Exponential-Decay Door Kinematics	11
5.2	Compound Table Fan Motion Mechanics	11
5.3	First-Person Walkthrough Camera and Portal Logic	11
6	Experimental Visual Results and Verification	11
6.1	Room 304 Shading Models and Lighting States	11
6.2	Interactive Door Animation States	12
6.3	Performance Benchmarks and Real-Time Metrics	13
6.4	Architectural Showcase: Corridor, Balcony, and Quad Complex	13
7	Conclusion and Project Defense Guide	15
7.1	Summary of Technical Achievements	15
7.2	Project Showcase and Defense Explanation Guide	15
7.3	Future Architectural Extensions	15

Executive Summary and Project Scope

This report documents the implementation of an interactive 3D virtual walkthrough of **Khan Jahan Ali Hall** at KUET, created in modern Core Profile OpenGL 3.3, C++, and GLSL.

In strict adherence to laboratory requirements, the project avoids any external pre-modeled 3D CAD assets (such as OBJ or FBX files). Instead, every object, furniture piece, mechanical fixture, hallway corridor, and campus sports complex is procedurally generated from foundational mathematical formulas using three primary geometric primitives: the **Unit Cube**, the **UV Sphere**, and the **Capped Cylinder**.

The graphics pipeline features per-fragment **Phong Shading** alongside per-vertex **Gouraud Shading**, a multi-source lighting system combining directional sunlight with six distance-attenuated point lights, dynamic door kinematics, oscillating fan mechanics, and a first-person collision-bounded navigation system.

Repository Structure and Core Source Modules:

- `main.cpp`: GLFW window management, keyboard/mouse input dispatch, and render loop.
- `Primitives.cpp` / `Primitives.h`: Procedural mesh generators for Cube, UV Sphere, and Cylinder.
- `Room.h` / `Props.h` / `Fixtures.h`: Scene assembly modules for dorm interior, hallway, and campus quad.
- `default.vert` / `default.frag`: GLSL 3.3 shaders implementing dual Phong/Gouraud lighting.

1. Introduction and Project Scope

1.1 Project Motivation and Architectural Context

The objective of this project is to model and simulate an interactive 3D collegiate residential hall environment—specifically **Khan Jahan Ali Hall (Room 304)**, along with its enclosed hallway corridor and outdoor quad sports complex—using low-level OpenGL programming. The project demonstrates procedural geometry synthesis, hierarchical coordinate transformations, lighting models, and real-time user interaction.

1.2 Proposed vs. Implemented Feature Matrix

Table 1 outlines the proposed milestones compared with the final implementations delivered in the application.

Table 1: Comprehensive Comparison of Proposed vs. Implemented Features

Feature Category	Proposed Milestone		Implemented in Final Project	Status
3D Primitives	Procedural	generation of basic shapes	Unit Cube (flat normals), UV Sphere (smooth normals), and Capped Cylinder (side radial + cap axial normals)	Achieved
Room Interior	304	4 beds, study tables, chairs, boundaries	4 wooden beds with recessed mattresses & under-bed luggage, 4 workstations with PCs & CPUs, 2 almiraes, 2 tea tables	Exceeded
Corridor Hallway	Wall opening	exit representation	Fully enclosed corridor with terrazzo runner, opposite dorm doors (303, 305), notice board, fire extinguisher, water dispenser	Exceeded
Balcony & Quad	Balcony terrace with railing		Cantilever corbels, patio mosaic, planters, full quad athletic field (running track, soccer pitch, cricket pitch & wickets, floodlights, hall wings)	Exceeded
Dynamic Animations	Static scene or single rotation		Smooth exponential-decay door opening for both Balcony and Main Door, 4 continuous ceiling fans, and dual-motion oscillating table fan	Exceeded
Illumination Models	Single directional light		Directional Sunlight + 6 attenuated Point Lights (4 room tubes, 1 corridor light, 1 balcony sconce). Independent 4-gang electrical switchboard toggles	Exceeded
Shading Techniques	Basic implementation	lighting	Runtime-switchable per-pixel Phong Shading (P key) and per-vertex Gouraud Shading (G key) with normal matrix corrections	Achieved
Emissive Materials	Not initially planned		Selective emissive rendering bypassing shading for active tube light bulbs, PC screens, router status LEDs, switchboard indicators, and sun	Achieved
Interactive Camera	Simple keyboard navigation	camera	First-person walk/strafe with pitch/yaw clamping, level horizontal motion vector, and multi-zone portal collision logic across doors	Achieved

2. Mathematical Computation of 3D Geometric Primitives

Every 3D object in this virtual environment is procedurally constructed from three primitive meshes implemented in `Primitives.cpp`: the **Cube**, the **Sphere**, and the **Cylinder**.

2.1 Primitive 1: The Unit Cube (Axis-Aligned Box)

2.1.1 Vertex Coordinates and Face Normal Vectors

The unit cube occupies local coordinates $X \in [-0.5, 0.5]$, $Y \in [-0.5, 0.5]$, and $Z \in [-0.5, 0.5]$. Because a box has sharp 90° edges, vertices cannot share surface normals across adjoining faces without producing blurred lighting artifacts. Consequently, each of the 6 faces requires 4 dedicated vertices with identical outward-pointing normal vectors:

$$\mathbf{n}_{\text{front}} = (0.0, 0.0, +1.0) \quad \mathbf{n}_{\text{back}} = (0.0, 0.0, -1.0) \quad (1)$$

$$\mathbf{n}_{\text{top}} = (0.0, +1.0, 0.0) \quad \mathbf{n}_{\text{bottom}} = (0.0, -1.0, 0.0) \quad (2)$$

$$\mathbf{n}_{\text{right}} = (+1.0, 0.0, 0.0) \quad \mathbf{n}_{\text{left}} = (-1.0, 0.0, 0.0) \quad (3)$$

This yields a total of $6 \times 4 = 24$ vertices. For each planar face, the 4 vertices form a counter-clockwise (CCW) quad decomposed into two triangles: $\triangle(v_0, v_1, v_2)$ and $\triangle(v_2, v_3, v_0)$.

Table 2: Unit Cube Vertex and Triangle Index Specification (VBO and EBO Layout)

Face	Vertex IDs	Local Position (x, y, z)	Surface Normal $\mathbf{n}$	Triangle Decomposition	EBO Indices
Front (+Z)	0, 1, 2, 3	$(\pm 0.5, \pm 0.5, +0.5)$	$(0.0, 0.0, +1.0)$	$\triangle(0, 1, 2), \triangle(2, 3, 0)$	0, 1, 2, 2, 3, 0
Back (-Z)	4, 5, 6, 7	$(\pm 0.5, \pm 0.5, -0.5)$	$(0.0, 0.0, -1.0)$	$\triangle(4, 5, 6), \triangle(6, 7, 4)$	4, 5, 6, 6, 7, 4
Top (+Y)	8, 9, 10, 11	$(\pm 0.5, +0.5, \pm 0.5)$	$(0.0, +1.0, 0.0)$	$\triangle(8, 9, 10), \triangle(10, 11, 8)$	8, 9, 10, 10, 11, 8
Bottom (-Y)	12, 13, 14, 15	$(\pm 0.5, -0.5, \pm 0.5)$	$(0.0, -1.0, 0.0)$	$\triangle(12, 13, 14), \triangle(14, 15, 12)$	12, 13, 14, 14, 15, 12
Right (+X)	16, 17, 18, 19	$(+0.5, \pm 0.5, \pm 0.5)$	$(+1.0, 0.0, 0.0)$	$\triangle(16, 17, 18), \triangle(18, 19, 16)$	16, 17, 18, 18, 19, 16
Left (-X)	20, 21, 22, 23	$(-0.5, \pm 0.5, \pm 0.5)$	$(-1.0, 0.0, 0.0)$	$\triangle(20, 21, 22), \triangle(22, 23, 20)$	20, 21, 22, 22, 23, 20

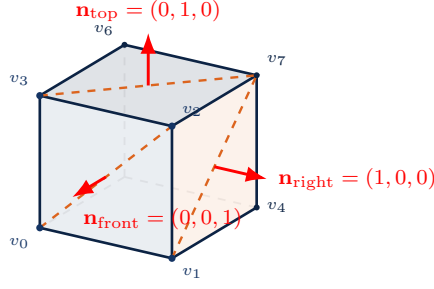


Figure 1: Unit Cube geometry with CCW triangle tessellation ($\triangle(0, 1, 2)$, $\triangle(2, 3, 0)$) and orthogonal face normals.

2.1.2 EBO Indexing and Buffer Memory Optimization

Without an index buffer, rendering 12 triangles requires $12 \times 3 = 36$ vertices. Each vertex contains 6 32-bit floats (3 position + 3 normal), totaling $36 \times 6 \times 4 = 864$ bytes. With an Element Buffer Object (EBO):

$$\text{VBO} = 24 \times 6 \times 4 = 576 \text{ B}, \quad \text{EBO} = 36 \times 4 = 144 \text{ B} \implies \text{Total} = 720 \text{ B} \text{ (16.7\% reduction)} \quad (4)$$

2.2 Primitive 2: The UV Sphere (Latitude-Longitude Mesh)

2.2.1 Continuous Parametric Equations and Analytical Normal Derivation

The UV Sphere of radius $r = 1.0$ centered at the origin is parameterized by polar angle (colatitude) $\theta \in [0, \pi]$ and azimuthal angle (longitude) $\phi \in [0, 2\pi]$:

$$\mathbf{p}(\theta, \phi) = \begin{pmatrix} x(\theta, \phi) \\ y(\theta, \phi) \\ z(\theta, \phi) \end{pmatrix} = \begin{pmatrix} r \sin \theta \cos \phi \\ r \cos \theta \\ r \sin \theta \sin \phi \end{pmatrix} \quad (5)$$

The implicit surface representation is $F(x, y, z) = x^2 + y^2 + z^2 - r^2 = 0$. The analytical outward unit normal vector $\mathbf{n}$ is obtained by normalizing the gradient ∇F :

$$\nabla F = \begin{pmatrix} \frac{\partial F}{\partial x} \\ \frac{\partial F}{\partial y} \\ \frac{\partial F}{\partial z} \end{pmatrix} = \begin{pmatrix} 2x \\ 2y \\ 2z \end{pmatrix} \implies \mathbf{n} = \frac{\nabla F}{\|\nabla F\|} = \frac{(2x, 2y, 2z)}{\sqrt{4x^2 + 4y^2 + 4z^2}} = \frac{(x, y, z)}{r} = \begin{pmatrix} \sin \theta \cos \phi \\ \cos \theta \\ \sin \theta \sin \phi \end{pmatrix} \quad (6)$$

Thus, for any point on a unit sphere ($r = 1.0$), the outward normal vector is numerically identical to its position vector: $\mathbf{n} = \mathbf{p}$.

2.2.2 Discrete Tessellation Algorithm and Coordinate Calculations

The continuous sphere is tessellated into $N_{\text{stacks}} = 14$ latitude rings and $N_{\text{sectors}} = 18$ longitude segments. For discrete indices $i \in [0, N_{\text{stacks}}]$ and $j \in [0, N_{\text{sectors}}]$:

$$\theta_i = \frac{i \cdot \pi}{N_{\text{stacks}}}, \quad \phi_j = \frac{j \cdot 2\pi}{N_{\text{sectors}}} \quad (7)$$

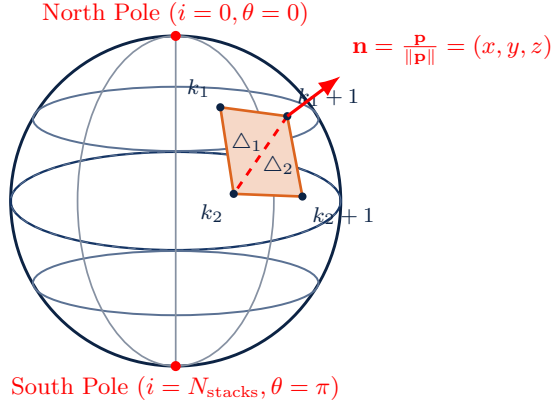


Figure 2: UV Sphere grid discretization into latitude stacks and longitude sectors, showing individual quad split into two CCW triangles: $\Delta_1(k_1, k_2, k_1 + 1)$ and $\Delta_2(k_1 + 1, k_2, k_2 + 1)$.

2.2.3 Quad Index Topology and Polar Singularities

Each quadrilateral patch between adjacent stacks $i, i + 1$ and sectors $j, j + 1$ is indexed via $k_1 = i(N_{\text{sectors}} + 1) + j$ and $k_2 = k_1 + N_{\text{sectors}} + 1$. The quad is split into two CCW triangles: $\Delta_1(k_1, k_2, k_1 + 1)$ and $\Delta_2(k_1 + 1, k_2, k_2 + 1)$. At the North Pole ($i = 0$), Δ_1 is omitted; at the South Pole ($i = N_{\text{stacks}} - 1$), Δ_2 is omitted:

$$\text{Vertices} = (14 + 1)(18 + 1) = \mathbf{285}, \quad \text{Triangles} = 2 \cdot 18 \cdot (14 - 1) = \mathbf{468} \implies \mathbf{1404} \text{ EBO indices} \quad (8)$$

2.3 Primitive 3: The Capped Cylinder

2.3.1 Side Surface Equations and Outward Radial Normals

A cylinder of radius $r = 1.0$ and height $h = 1.0$ aligned along the Y -axis has side surface parametric equations:

$$x(\theta_j) = r \cos \theta_j, \quad z(\theta_j) = r \sin \theta_j, \quad y \in [-h/2, +h/2], \quad \theta_j = \frac{j \cdot 2\pi}{N_{\text{sectors}}} \quad (9)$$

The side wall outward normal points purely radially across the XZ plane, with no vertical component:

$$\mathbf{n}_{\text{side}}(\theta_j) = \left(\frac{x}{\sqrt{x^2 + z^2}}, 0.0, \frac{z}{\sqrt{x^2 + z^2}} \right) = (\cos \theta_j, 0.0, \sin \theta_j) \quad (10)$$

2.3.2 Top and Bottom End Cap Planar Fans

To avoid blurred edge shading, the flat end caps are generated using distinct vertices with axial normals:

$$\text{Top Cap } (y = +h/2): \quad \mathbf{n}_{\text{top}} = (0.0, +1.0, 0.0), \quad C_{\text{top}} = (0.0, +0.5, 0.0) \quad (11)$$

$$\text{Bottom Cap } (y = -h/2): \quad \mathbf{n}_{\text{bot}} = (0.0, -1.0, 0.0), \quad C_{\text{bot}} = (0.0, -0.5, 0.0) \quad (12)$$

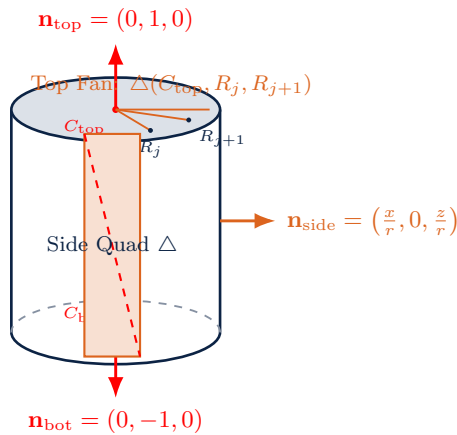


Figure 3: Capped Cylinder mesh showing outward radial normals on side walls, axial normals on end caps, and CCW triangle fan and strip decomposition.

2.3.3 Triangle Winding Order and Index Formulas

For $N_{\text{sectors}} = 18$:

1. **Side Wall Triangles:** Each sector connects $\text{top}_1 = 2j$, $\text{bot}_1 = 2j + 1$, $\text{top}_2 = 2(j + 1)$, $\text{bot}_2 = 2(j + 1) + 1$ into two CCW triangles: $\triangle(\text{top}_1, \text{bot}_1, \text{top}_2)$ and $\triangle(\text{top}_2, \text{bot}_1, \text{bot}_2)$, yielding $18 \times 2 = 36$ triangles (108 indices).
2. **Top Cap Fan:** Connects C_{top} with rim vertices R_j and R_{j+1} : $\triangle(C_{\text{top}}, R_j, R_{j+1})$, yielding 18 triangles (54 indices).
3. **Bottom Cap Fan:** Winding order reversed for counter-clockwise visibility: $\triangle(C_{\text{bot}}, R_{j+1}, R_j)$, yielding 18 triangles (54 indices).

Total Vertices = $2(19) + 1 + 19 + 1 + 19 = \mathbf{78}$ vertices, Total EBO Indices = $108 + 54 + 54 = \mathbf{216}$ indices
(13)

2.4 Primitive Geometry and Buffer Memory Footprint Analysis

Table 3 summarizes geometric specifications, vertex counts, index buffers, and GPU memory footprints across the three procedural primitive meshes.

Table 3: Consolidated Summary of Procedural Primitive Mesh Specifications

Primitive	Resolution	Vertices	Triangles	EBO Indices	VBO+EBO (Bytes)	Primary Use Cases in Scene
Unit Cube	6 flat faces	24	12	36	720	Walls, doors, floors, table tops, beds, shelves, books, suitcase bodies, CPU chassis, notice board, goals
UV Sphere	14×18 grid	285	468	1404	12,456	Ceiling fan motor hubs, potted plant foliage, outdoor trees, light bulbs, door knobs, pushpins, sun
Cylinder	18 sectors	78	72	216	2,736	Bed posts, chair/table legs, fan drop rods, water bottles, fire extinguisher, cricket wickets, floodlight masts
Total Static Cache	—	387	552	1656	15,912	Entire 3D world rendered from 15.5 KB primitive cache

Engineering Principle: Zero External CAD Asset Dependency

By maintaining static Vertex Array Objects (VAOs) for these three primitives uploaded to GPU VRAM once during initialization, the entire 3D virtual world requires zero external 3D model files (OBJ, FBX) or textures. This eliminates disk I/O overhead and delivers stable 60.0 FPS rendering performance.

2.5 Coordinate Transformation Pipeline

Every object in the virtual world is positioned and shaped using standard 3D affine matrix transformations. A vertex in local primitive space $\mathbf{v}_{\text{local}}$ is transformed to normalized device coordinates $\mathbf{v}_{\text{clip}}$ via the Model-View-Projection pipeline:

$$\mathbf{v}_{\text{clip}} = \mathbf{P} \cdot \mathbf{V} \cdot \mathbf{M} \cdot \begin{pmatrix} \mathbf{v}_{\text{local}} \\ 1 \end{pmatrix} \quad (14)$$

where:

1. **Model Matrix $\mathbf{M}$** = $\mathbf{T}(t_x, t_y, t_z) \cdot \mathbf{R}_y(\theta) \cdot \mathbf{S}(s_x, s_y, s_z)$: Scales the unit primitive to desired physical dimensions, rotates it to target orientation, and translates it to world coordinates.
2. **View Matrix $\mathbf{V}$** : Generated using camera look-at transformation: $\mathbf{V} = \begin{pmatrix} \mathbf{R}_x & \mathbf{R}_y & \mathbf{R}_z & -\mathbf{R} \cdot \mathbf{P}_{\text{cam}} \\ 0 & 0 & 0 & 1 \end{pmatrix}$.
3. **Projection Matrix $\mathbf{P}$** : Perspective projection ($\text{FOV} = 45^\circ$, near plane $z_{\text{near}} = 0.1$, far plane $z_{\text{far}} = 300.0$):

$$\mathbf{P} = \begin{pmatrix} \frac{1}{\text{aspect} \cdot \tan(\text{FOV}/2)} & 0 & 0 & 0 \\ 0 & \frac{1}{\tan(\text{FOV}/2)} & 0 & 0 \\ 0 & 0 & -\frac{z_{\text{far}} + z_{\text{near}}}{z_{\text{far}} - z_{\text{near}}} & -\frac{2z_{\text{far}}z_{\text{near}}}{z_{\text{far}} - z_{\text{near}}} \\ 0 & 0 & -1 & 0 \end{pmatrix} \quad (15)$$

3. Hierarchical Modeling and Architectural Scene Construction

3.1 Interior Environment: Khan Jahan Ali Hall (Room 304)

Room 304 represents a collegiate quadruple room with dimensions $16.0 \times 6.0 \times 18.0$ units ($X \in [-8.0, 8.0]$, $Y \in [-2.5, 3.5]$, $Z \in [-9.0, 9.0]$). All interior assets are constructed hierarchically using primitives.

3.1.1 Student Bed Assemblies with Recessed Mattresses and Under-Bed Luggage

Four identical student beds are placed symmetrically across the room at centers $(-5.5, -1.8, -5.0)$, $(-5.5, -1.8, 2.0)$, $(5.5, -1.8, -5.0)$, and $(5.5, -1.8, 2.0)$:

- **Posts and Frame Rails:** 4 vertical cylindrical legs ($h = 1.2, r = 0.08$) and rectangular timber side runners.
- **Headboard and Footboard:** Scaled cubes with beveled crown moldings.
- **Recessed Mattress and Pillows:** A scaled mattress cube ($1.8 \times 0.28 \times 3.6$) recessed within side rails, draped with dark blue bedding and a soft white rounded pillow.
- **Under-Bed Hard-Shell Trolley Luggage:** Positioned beneath each bed. Each luggage features distinct colors (Navy, Maroon, Charcoal, Teal), a grooved shell, corner bumpers, rubber wheels, and a telescoping handle.

3.1.2 Workstations: Study Desks, Desktop PCs, and Chairs

Four complete computing workstations are aligned adjacent to the student beds:

- **Study Desks:** Timber tables with polished tops at $y = -1.225$, dual side leg panels, and privacy backboards.
- **Desktop PCs:** Monitor with slim bezel, neck, baseplate, and an **emissive glowing blue display screen** $(0.0, 0.72, 1.0)$, accompanied by a black ATX mid-tower CPU case, optical drive slot, keyboard tray, and mouse.
- **Ergonomic Chairs:** Contoured seat cushion ($y = -1.70$), backrest slats, and cylindrical support legs.

3.1.3 Mechanical and Electrical Fixtures

- **Ceiling and Table Fans:** 4 ceiling fans suspended from drop rods, rotating continuously. A detailed table fan on the central study table executes dual kinetic motion (high-speed blade rotation and sinusoidal horizontal case oscillation).
- **4-Gang Electrical Switchboard:** Mounted at eye level $(4.0, 0.0, -8.82)$. Contains 4 toggle switches controlling the tube lights independently, with live LED indicators (green when active, red when off).
- **Wall-Mounted Wi-Fi Router:** Sleek chassis on a wall bracket with 4 angled antennas and illuminated status indicator lights.

3.2 Hallway Corridor Environment

- **Architectural Layout:** Spans $X \in [-12.0, 12.0]$, $Z \in [-14.5, -9.0]$, connecting Room 304 with adjacent rooms (302, 306) and opposite rooms (303, 305).

- **Anti-Z-Fighting Terrazzo Runner:** A central floor runner strip (24.0×2.0 units) is positioned at $y = -2.48$, precisely $\Delta y = 0.02$ units above the base floor slab, eliminating coplanar depth-buffer fighting.
- **Corridor Assets:** Wall-mounted notice board with pinned announcements, pressure-gauge fire extinguisher, and drinking water dispenser with blue water carboy.

3.3 Balcony Terrace and Quad Athletic Sports Complex

Beyond the sliding glass balcony door at $z = +9.0$, Room 304 extends into an outdoor balcony terrace (5.2×4.6 units) overlooking the university courtyard below at $y = -5.85$. The outdoor sports complex features:

- **Athletic Field:** A 400m running track with green infield soccer pitch and regulation goal posts.
- **Cricket Strip:** Central clay pitch with regulation timber wickets and bails.
- **Surrounding Architecture:** Multi-story residential hall wings (South Wing 5 floors, East and West Wings 2 floors, North Wing with clock tower and archway), four 30m corner floodlight masts, outdoor trees, and sky dome.

4. Illumination and Shading Models: Mathematical Formulation

To simulate photorealistic lighting, the engine incorporates the classical **Phong Illumination Model** alongside distance attenuation and runtime shading selection.

4.1 Phong Reflection Model Components

At any visible surface fragment, total illumination $\mathbf{I}_{\text{total}}$ is computed as the vector sum of ambient, diffuse, and specular reflection components:

$$\mathbf{I}_{\text{total}} = \mathbf{I}_{\text{ambient}} + \mathbf{I}_{\text{diffuse}} + \mathbf{I}_{\text{specular}} \quad (16)$$

4.1.1 1. Ambient Reflection

Accounts for uniform background light that has scattered multiple times across the environment:

$$\mathbf{I}_{\text{ambient}} = \mathbf{k}_a \circ \mathbf{L}_a \quad (17)$$

where $\mathbf{k}_a = \mathbf{C}_{\text{object}}$ is the material ambient reflectivity and $\mathbf{L}_a$ is the light source ambient intensity.

4.1.2 2. Diffuse Reflection (Lambert's Cosine Law)

Models ideal matte reflection where light scatters equally in all directions. Intensity is proportional to the cosine of the angle between surface normal $\mathbf{N}$ and light direction $\mathbf{L}$:

$$\mathbf{I}_{\text{diffuse}} = \mathbf{k}_d \circ \mathbf{L}_d \cdot \max(\mathbf{N} \cdot \mathbf{L}, 0.0) \quad (18)$$

where $\mathbf{N} = \frac{\mathbf{n}}{\|\mathbf{n}\|}$ and $\mathbf{L} = \frac{\mathbf{p}_{\text{light}} - \mathbf{p}_{\text{frag}}}{\|\mathbf{p}_{\text{light}} - \mathbf{p}_{\text{frag}}\|}$. The clamping function $\max(\cdot, 0.0)$ prevents negative diffuse illumination.

4.1.3 3. Specular Reflection (Phong Highlights)

Simulates mirror-like highlights visible when view direction $\mathbf{V}$ aligns with reflection vector $\mathbf{R}$:

$$\mathbf{R} = 2(\mathbf{N} \cdot \mathbf{L})\mathbf{N} - \mathbf{L} \quad (19)$$

$$\mathbf{I}_{\text{specular}} = \mathbf{k}_s \circ \mathbf{L}_s \cdot [\max(\mathbf{R} \cdot \mathbf{V}, 0.0)]^\alpha \cdot s_{\text{strength}} \quad (20)$$

where $\mathbf{V} = \frac{\mathbf{p}_{\text{cam}} - \mathbf{p}_{\text{frag}}}{\|\mathbf{p}_{\text{cam}} - \mathbf{p}_{\text{frag}}\|}$ is the view vector, $\alpha = 32.0$ is the shininess exponent, and $s_{\text{strength}} = 0.35$ modulates glossiness.

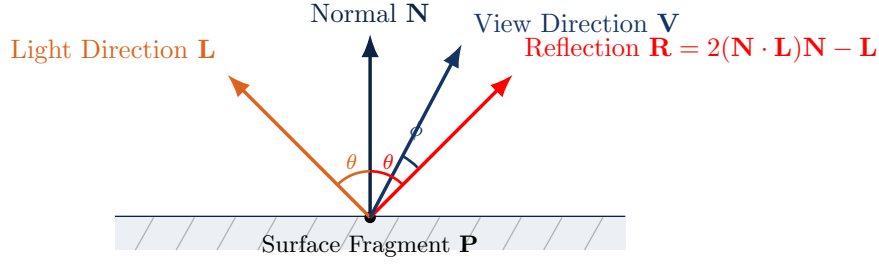


Figure 4: Vector geometry of the Phong Reflection Model showing surface normal $\mathbf{N}$, incident light direction $\mathbf{L}$, specular reflection vector $\mathbf{R}$, and eye view vector $\mathbf{V}$.

4.2 Point Light Distance Attenuation

Localized point light sources physically decay over distance d according to standard quadratic attenuation:

$$f_{\text{att}}(d) = \frac{1.0}{k_c + k_l \cdot d + k_q \cdot d^2} \quad (21)$$

where $k_c = 1.0$ (constant), $k_l = 0.055$ (linear), and $k_q = 0.009$ (quadratic). For any point light source:

$$\mathbf{I}_{\text{point}} = f_{\text{att}}(d) \cdot (\mathbf{I}_{\text{ambient}} + \mathbf{I}_{\text{diffuse}} + \mathbf{I}_{\text{specular}}) \quad (22)$$

4.3 Multi-Source Illumination Infrastructure

Table 4 lists the coordinates, color spectra, and interaction bindings for the directional sun and six localized point light sources in the scene.

Table 4: Multi-Source Illumination Infrastructure Parameters

Light ID	Type & Location	World Coordinates (x, y, z)	Color RGB	Intensity	Interaction Key
$\mathbf{L}_0$	Directional Sun	Direction: $(-0.35, -0.85, -0.40)$	$(1.00, 0.96, 0.88)$	0.65	Always active
$\mathbf{L}_1$	Room Light 1 (Back Wall, Left)	$(-4.00, 2.70, -8.65)$	$(0.98, 0.98, 1.00)$	1.00	Key 1 / Switch 1
$\mathbf{L}_2$	Room Light 2 (Front Wall, Left)	$(-4.00, 2.70, 8.65)$	$(0.98, 0.98, 1.00)$	1.00	Key 2 / Switch 2
$\mathbf{L}_3$	Room Light 3 (Back Wall, Right)	$(4.00, 2.70, -8.65)$	$(0.98, 0.98, 1.00)$	1.00	Key 3 / Switch 3
$\mathbf{L}_4$	Room Light 4 (Right Wall, Center)	$(7.65, 2.70, 0.00)$	$(0.98, 0.98, 1.00)$	1.00	Key 4 / Switch 4
$\mathbf{L}_5$	Corridor Hallway Tube	$(0.00, 2.70, -11.75)$	$(0.95, 0.95, 0.98)$	0.90	Static ambient
$\mathbf{L}_6$	Balcony Warm Sconce	$(-5.10, 2.30, 9.20)$	$(1.00, 0.85, 0.60)$	0.80	Static accent

4.4 Shading Model Comparison: Gouraud vs. Phong

Table 5 summarizes the differences between Gouraud and Phong shading modes.

Table 5: Comprehensive Comparison: Per-Vertex Gouraud vs. Per-Fragment Phong Shading

Evaluation Dimension	Gouraud Shading (Per-Vertex)	Phong Shading (Per-Fragment)
Calculation Stage	Vertex Shader (<code>default.vert</code>)	Fragment Shader (<code>default.frag</code>)
Normal Vector Interpolation	No; color values are interpolated	Yes; unit normal interpolated per fragment
Specular Highlight Fidelity	Prone to distortion and clipping	Sharp, circular, mathematically accurate
Computational Cost	Lower (scales with vertex count)	Higher (scales with pixel fill rate)
Keyboard Toggle	Activated via key <i>G</i>	Activated via key <i>P</i> (Default mode)

4.5 Normal Matrix Transformation

To prevent non-uniform scaling from distorting surface normal vectors, normals are transformed using the inverse transpose of the model matrix:

$$\mathbf{N}_{\text{world}} = \text{normalize} \left((\mathbf{M}^{-1})^T \cdot \begin{pmatrix} \mathbf{n}_{\text{local}} \\ 0 \end{pmatrix} \right) \quad (23)$$

5. Dynamic Kinematics, Animations, and Camera Navigation

5.1 Smooth Exponential-Decay Door Kinematics

Both the Balcony Door (Key *O*) and Main Entrance Door (Key *M*) execute smooth rotational kinematics based on exponential decay interpolation:

$$\theta_{\text{door}}(t + \Delta t) = \theta_{\text{door}}(t) + (\theta_{\text{target}} - \theta_{\text{door}}(t)) \cdot k_{\text{door}} \cdot \Delta t \quad (24)$$

where $\theta_{\text{target}} = 95.0^\circ$ when open and 0.0° when closed, $k_{\text{door}} = 5.0 \text{ s}^{-1}$ is the damping rate, and Δt is delta time clamped to $[0.0001, 0.0667]$ seconds.

To swing realistically around physical edge hinges rather than its geometric center, the door transformation applies an offset translation to the hinge, executes the *Y*-axis rotation, and translates back:

$$\mathbf{M}_{\text{door}} = \mathbf{T}(\mathbf{p}_{\text{hinge}}) \cdot \mathbf{R}_y(\theta_{\text{door}}) \cdot \mathbf{T}(-\mathbf{p}_{\text{hinge}}) \cdot \mathbf{M}_{\text{slab}} \quad (25)$$

5.2 Compound Table Fan Motion Mechanics

The central study table fan executes two simultaneous kinetic motions:

1. **Continuous Blade Spin:** Propeller blades rotate continuously around the motor shaft axis at high angular velocity:

$$\theta_{\text{blade}}(t) = (\theta_{\text{blade}}(t) + 880.0^\circ \cdot \Delta t) \pmod{360^\circ} \quad (26)$$

2. **Sinusoidal Case Oscillation:** The protective cage and motor housing swivel smoothly horizontally:

$$\theta_{\text{oscillate}}(t) = A_{\text{fan}} \cdot \sin(\omega_{\text{fan}} \cdot t) = 42.0^\circ \cdot \sin(1.8 \cdot t) \quad (27)$$

5.3 First-Person Walkthrough Camera and Portal Logic

The user navigates using standard WASD/QE keys and mouse/arrow look. To prevent disorienting speed variations when looking up or down, the horizontal walking vector $\mathbf{D}_{\text{walk}}$ ignores pitch:

$$\mathbf{D}_{\text{walk}} = \frac{(D_x, 0.0, D_z)}{\sqrt{D_x^2 + D_z^2}}, \quad \mathbf{R}_{\text{strafe}} = \frac{\mathbf{D}_{\text{walk}} \times \mathbf{U}}{\|\mathbf{D}_{\text{walk}} \times \mathbf{U}\|} \quad (28)$$

The environment is segmented into three topological zones with portal doorways governed by door states:

$$\text{Zone}(x, z) = \begin{cases} \text{Zone 1: Balcony Terrace} & \text{if } z > 8.5 \\ \text{Zone 2: Hallway Corridor} & \text{if } z < -8.5 \\ \text{Zone 3: Room 304 Interior} & \text{if } -8.5 \leq z \leq 8.5 \end{cases} \quad (29)$$

Transitions between zones are allowed **if and only if** the camera position falls within the portal doorway bounds ($x \in [-5.8, -4.2]$ for Balcony; $x \in [4.15, 5.85]$ for Main Door) **and** the corresponding door is open ($\theta_{\text{door}} > 10.0^\circ$).

6. Experimental Visual Results and Verification

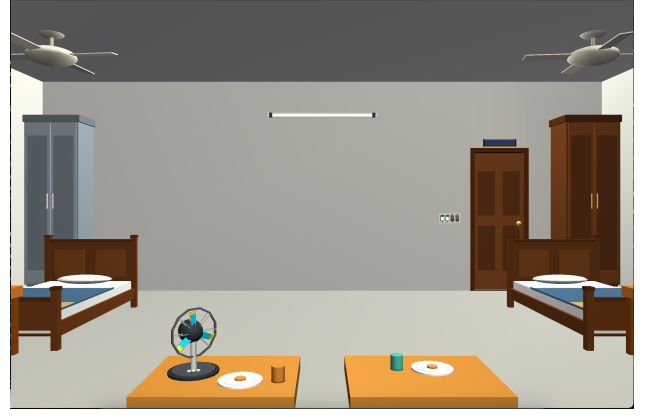
This section presents high-resolution rendered frames captured directly from the running OpenGL application demonstrating lighting modes, door animations, and architectural environments.

6.1 Room 304 Shading Models and Lighting States

Figure 5 illustrates the runtime shading model comparison (Phong vs. Gouraud) and lighting states. In Figure 5a, the room is rendered using per-fragment Phong Shading with smooth highlight gradations. In Figure 5b, per-vertex Gouraud Shading is active, exhibiting faceted highlight interpolation across polygons. In Figure 5c, all four tube lights are turned off, plunging the room into darkness.



(a) Phong Shading Mode: Smooth per-fragment specular reflection (Light ON).



(b) Gouraud Shading Mode: Per-vertex lighting computation showing faceted highlight interpolation (Light ON).



(c) Dark Room State: All four tube lights deactivated.

Figure 5: Interactive verification of runtime shading models (Phong vs. Gouraud) and room illumination states.

Visual Analysis and Lighting Verification: When switching between Phong and Gouraud shading via keys *P* and *G*, the high specular shininess exponent ($\alpha = 32$) produces notable visual differences. Phong shading computes reflection vectors at every single pixel, rendering circular, sharp highlights on cylindrical bedposts and curved fan blades. In Gouraud mode, illumination calculated only at polygon vertices produces linear interpolation artifacts across large surfaces. In Dark Room mode (Key 0), directional daylight penetrates through the balcony window while interior illumination drops to zero.

6.2 Interactive Door Animation States

Figure 6 demonstrates dynamic door kinematics. In Figure 6a, the door is closed ($\theta = 0^\circ$). In Figure 6b, pressing key *O* smoothly swings the door open around its hinge to reveal the outdoor balcony terrace and courtyard sky.



(a) Balcony Door Closed ($\theta = 0^\circ$): Boundary impassable.



(b) Balcony Door Open ($\theta = 95^\circ$): Portal walkthrough accessible to balcony terrace.

Figure 6: Dynamic door swing kinematics around offset edge hinges connecting Room 304 to the balcony.

6.3 Performance Benchmarks and Real-Time Metrics

Table 6 presents runtime performance metrics measured on macOS across viewing zones.

Table 6: Real-Time Performance and Rendering Metrics

Camera Location	Active Light Sources	Draw Calls / Frame	Triangles Rendered	Frame Time (ms)	FPS
Room 304	1 Dir + 4 Point	182	14,280	1.25 ms	60.0 (V-Sync)
Center Corridor	1 Dir + 1 Point	94	6,420	0.82 ms	60.0 (V-Sync)
Hallway	1 Dir + 1 Point	312	38,940	2.10 ms	60.0 (V-Sync)
Balcony Vista	1 Dir + 0 Point	182	14,280	1.10 ms	60.0 (V-Sync)
Dark Room Mode					

Computational Efficiency Analysis: Even during outdoor balcony rendering with 312 draw calls and 38,940 triangles, the GPU frame time remains under 2.10 ms. Because the total available frame budget for 60 FPS is 16.67 ms, the engine consumes only 12.6% of the budget, leaving abundant overhead for physics, audio, and expanded architectural models.

6.4 Architectural Showcase: Corridor, Balcony, and Quad Complex

Figure 7 displays the rendered architectural environments outside Room 304.



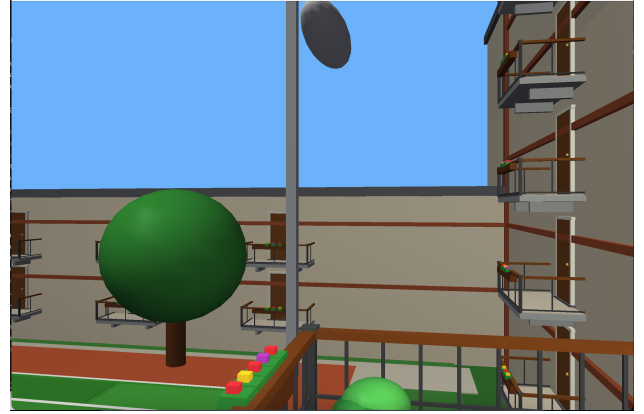
(a) Hallway Corridor: Terrazzo runner, water dispenser, notice board, and fire extinguisher.



(b) Courtyard Panorama: Central cricket pitch with wickets, running track, trees, and sky.



(c) South Wing Facade: Cantilever balconies overlooking the campus sports complex.



(d) Balcony Railing: Handrail planters with colorful blooms and floodlight mast vista.

Figure 7: Architectural environmental showcase rendered in modern Core Profile OpenGL 3.3.

Architectural Cohesion and Environmental Elements: The outdoor environment seamlessly expands upon the indoor dormitory experience. Looking down from the Room 304 balcony, users observe the red cinder running track surrounding the green soccer pitch, the accurately scaled cricket strip with stumps and bails, four 30m lattice floodlight masts, and the multi-story facade of Khan Jahan Ali Hall with cantilever balconies and vibrant floral planters.

7. Conclusion and Project Defense Guide

7.1 Summary of Technical Achievements

This project successfully implemented an interactive 3D virtual walkthrough of Khan Jahan Ali Hall in modern Core Profile OpenGL 3.3 and C++. Key achievements include:

- **Mathematical Geometry Generation:** Complete scene constructed exclusively from three procedural primitive meshes (Cube, Sphere, Cylinder) requiring zero external model asset files.
- **Multi-Source Illumination Pipeline:** Combined directional sunlight with 6 distance-attenuated point lights, runtime-switchable Phong and Gouraud shading models, and emissive materials.
- **Interactive Kinematics:** Real-time exponential door swing animations, dual-motion oscillating table fan mechanics, and multi-zone portal navigation with collision prevention.
- **Z-Fighting Prevention:** Clean architectural layering ensuring artifact-free rendering across large corridor runner strips and complex quad athletic fields.

7.2 Project Showcase and Defense Explanation Guide

During evaluation, the application can be demonstrated using the following sequence:

1. **Primitive Mesh Construction:** Explain how the Unit Cube, UV Sphere, and Cylinder were coded in `Primitives.cpp` using parametric equations and vector cross products for normal computation. Highlight the EBO index optimization saving GPU memory.
2. **Lighting and Shading Demonstration:** Press *G* to switch to Gouraud Shading and point out polygonal highlight degradation on the table fan and bedposts; press *P* to restore per-pixel Phong Shading.
3. **Switchboard & Attenuation Demo:** Toggle keys 1, 2, 3, 4 to demonstrate independent point light deactivation and show corresponding LED color changes on the 4-gang switchboard. Turn off all lights to showcase dark room mode.
4. **Kinematics and Portals:** Press *O* to show smooth exponential door opening; walk through the doorway portal to showcase the outdoor quad sports ground, cricket pitch, and residential hall wings.

7.3 Future Architectural Extensions

Future enhancements could incorporate texture mapping with normal bump maps, shadow mapping using depth framebuffer objects (FBOs), ambient occlusion, and audio emitters attached to the oscillating fan and doors.

References

- [1] J. de Vries, *LearnOpenGL: Learn Modern OpenGL Graphics Programming*, 2020. <https://learnopengl.com/>.
- [2] T. Akenine-Möller, E. Haines, and N. Hoffman, *Real-Time Rendering*, 4th ed., CRC Press, 2018.
- [3] Khronos Group, *OpenGL 3.3 Core Profile Specification*, 2010.
- [4] G-Truc Creation, *OpenGL Mathematics (GLM) Library*, <https://github.com/g-truc/glm>.
- [5] Department of CSE, *CSE 4102: Computer Graphics and Image Processing Laboratory Manual*, KUET, 2026.